

Steam Turbine-Generator - O&M Manual

3-cylinder (HP/IP/LP) reheat turbine + 2-pole generator

Field	Value
Document No.	TG-OM-1100
Revision	Rev 3.1
Equipment OEM	(synthetic; BHEL/Siemens class)
Applicable standard	ISO 10816 / IEC / CEA norms
Classification	O&M reference - SYNTHETIC (Ballast demo)
Owner	Plant O&M / Reliability

Disclaimer: This is a synthetic document generated for the Ballast demo. Values are grounded in public standards and typical industry practice but do not represent any specific OEM's proprietary manual.

1. Vibration (ISO 10816-2, large turbine-generator sets)

For large turbo sets, both bearing-housing velocity and relative shaft vibration (proximity probes, displacement in micrometres) are monitored. Bearing-housing guide values below.

Condition	Shaft vib (um, pk-pk)	Bearing housing (mm/s)	Action
Good	< 90	< 2.8	Normal
Acceptable	90 - 165	2.8 - 5.3	Unrestricted
Alarm	165 - 240	5.3 - 8.5	Investigate
Danger/Trip	> 240	> 8.5	Trip protection

2. Key Process Limits

Point	Normal	Alarm	Trip
Main steam temp	537 degC	566 degC	571 degC
HP/IP metal temp (rotor)	<= 540 degC	566 degC	571 degC
Thrust bearing metal temp	70 - 100 degC	110 degC	120 degC
Lube oil pressure	9 - 10 bar	7 bar (low)	5 bar (trip)
Overspeed	3000 rpm	-	3300 rpm
Condenser vacuum	660 - 700 mmHg	600 (low)	550 (trip)

3. Service Intervals & Failure Modes

Task	Interval
Lube oil analysis	Monthly
Bearing inspection	Major overhaul (~ 4 yrs)
Governor/valve testing	Quarterly
Overspeed trip test	Post-overhaul / annual

- High vibration - imbalance, rub, misalignment, bearing wipe (lube failure).
- Rising thrust-bearing temp - thrust wear; risk of axial contact.
- Falling condenser vacuum - air ingress / CW fouling; derates output.